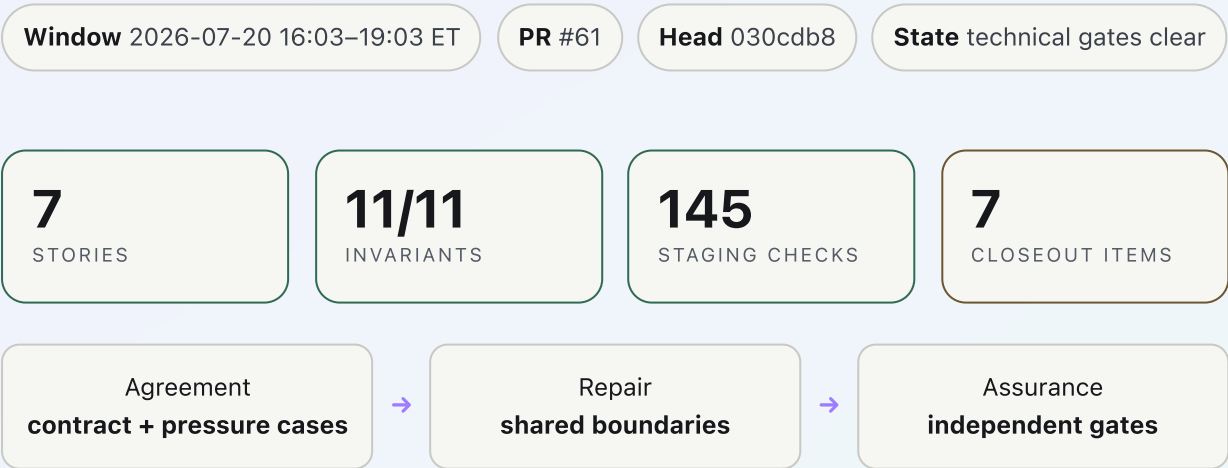


Agent UI Ultragoal retrospective

What shipped, what independent gates caught, what remains operationally open, and how the next targeted run should improve.



Sonik Agent UI Ultragoal retrospective

Final evidence package · G001–G007

Run window: 2026-07-20 16:03–19:03 America/New_York, plus final closeout **Product/review head:**

`030cdb86eda3aedba326392f1ad4e363cf960b44` **Branch:** `fix/dev-workbench-embedded-context-20260720` **Pull request:** [SonikFM/sonik-agent-ui#61](#) **Disposition:** implementation and local quality gates clear; push, remote rereview, CI on the new head, live authenticated operator proof, and durable Ultragoal closeout remain.

This is a retrospective, not a promotion claim. The final product head passed the recorded code and architecture gates, but PR 61 still points to the old head and the durable Ultragoal ledger has not yet been reconciled. Completed engineering evidence and remaining operational work are separated below.

1. Executive result

The run converted a broad reliability and UX completion brief into an agreement-first contract, repaired the highest-risk PR 61 boundaries, made session and telemetry persistence joinable and tenant-scoped, aligned the theme and JSON-render paths with the product authority, performed bounded staging-only reliability testing, and closed two independent review blockers through appended stories G006 and G007.

At product head `030cdb8`:

- independent code review recommends **APPROVE**;
- independent architecture review reports **CLEAR** with all 11 invariants passing;
- the changed-file AI-slop cleaner reports **PASS / no-op**;
- full typecheck, both production builds, focused telemetry/session tests, `git diff --check`, and the dependency-diff gate are recorded green;
- the current Agent UI Worker and Dev Workbench deployments supersede the pre-G007 deployments; and
- no dependency, oRPC stack, bespoke transport, second design system, or per-caller telemetry workaround was introduced.

The run is not fully closed operationally. PR 61 remains open and blocked at remote head `036f713`, with the old `CHANGES_REQUESTED` decision and green old-head CI. Authenticated live navigation/reload, protected Workbench login, embedded Booking host context, and downstream workflow recovery still require an operator with the relevant credentials and host environment.

2. Outcome matrix

AREA	FINAL EVIDENCE	STATUS	HONEST BOUNDARY
Agreement and acceptance baseline	Markdown/HTML/PDF contract, six pressure-case groups, ownership, gates, rollback	Complete	Baseline was a contract, not a completion claim.
PR 61 contract hardening	Eight unresolved review threads inventoried; contract suite moved from 0/6 to 8/8	Complete locally	Remote PR has not received the new head.
Session and prompt durability	Joinable message persistence; generation waits for the newest user message; owner-scoped hydration	Complete in repository evidence	Live authenticated reload/navigation remains an operator check.
Stream lifecycle telemetry	Request-aware spec/tool/terminal writers persist through the owning cloud adapter	Complete in repository evidence	Telemetry is intentionally fail-soft.
Telemetry privacy	Shared recursive sanitizer removes reasoning/thinking/scratchpad/chain-of-thought key variants before logs or persistence	Complete	Absence from the durable store is not proof an event never occurred.
Host/sandbox authority	Opaque host authority and Cloudflare control-plane secrets stay out of guest sandbox paths	Complete in contract/tests	Environment-specific credentials remain operator-owned.
Theme authority	DaisyUI/Tailwind and Sonik theme registry remain authoritative; Callout and reduced-motion issues repaired	Complete	Integrated visual/accessibility operator sweep remains useful follow-up.
JSON-render	Core/Svelte runtime, 33-component Sonik registry, artifact validation/promotion, JSONL, and guarded devtools	Complete for primary path	YAML wire and donor breadth remain partial; secondary renderers remain out of scope.
API reliability	Staging GET-only Pipe B health run: 145/145 checks, 0% failures, p95 113.947 ms	Complete for bounded liveness	Not workflow recovery, auth correctness, or write-path proof.
Independent quality gates	Cleaner PASS, code review APPROVE, architecture CLEAR, 11/11 invariants PASS	Complete at 030cdb8	Remote GitHub review decision remains old-head CHANGES_REQUESTED
Deployment receipts	Agent UI Worker version and protected Workbench URLs recorded	Complete as deployment receipts	Authenticated end-to-end operator journey is not yet proved.
PR and durable goal close	Push, new-head CI/rereview, G005-G007 ledger reconciliation, aggregate quality gate	Remaining	Leader-owned closeout only.

3. Executed Ultragoal and checkpoints

The run began from `.omx/ultragoal/brief.md` at 2026-07-20T20:03:01Z. The generated 18-goal draft was reduced to the five required stories before execution. Two review blockers then appended G006 and G007 instead of allowing a false completion claim.

G001 — Completion contract and PDF baseline

Started: 20:06:17Z · **Completed:** 20:24:27Z

- Produced `sonik-agent-ui-completion-contract-2026-07-20.{md,html,pdf}`.
- Classified capabilities as complete, partial, skeleton, or missing.
- Defined ownership, acceptance, deployment and rollback gates, theme authority, JSON-render boundaries, and six pressure-case groups.
- Validated a 12-page PDF and ran full tests/typecheck without changing product source.

G002 — PR 61 CodeRabbit and contract hardening

Started: 20:24:48Z · **Completed:** 20:54:38Z

- Inventoried eight unresolved PR comments and preserved Zod 4.3.6 as schema authority.
- Added no oRPC or dependency.
- Moved the bounded contract suite from RED 0/6 to GREEN 8/8.
- Prevented guest exposure of host authority and Cloudflare control-plane secrets; made Pipe B and preview/evidence truth explicit.
- Independent rereview approved the repaired head.

G003 — Live session, prompt, stream, telemetry, and host-context proof

Started: 20:54:45Z · **Completed:** 21:08:42Z

- Closed the immediate-submit race with joinable message persistence.
- Proved telemetry POST-to-session GET, cloud SQL ordering/isolation, secret redaction, durable correlation, and workspace-store regressions.
- Preserved session/run identity across the repository hydration contract.
- Full typecheck passed with one pre-existing `JsonRenderDevtools.svelte` initial-state warning.

G004 — Theme, component, and JSON-render audit

Started: 21:08:58Z · **Completed:** 21:32:28Z

- Kept DaisyUI/Tailwind and Sonik semantic tokens as the single design-system authority.
- Replaced the directional Callout stripe with a semantic full-border alert surface.
- Replaced gradient-clipped shimmer text with an opacity pulse and readable reduced-motion fallback.
- Passed `explorerCatalog` into dev-only JSON-render devtools while preserving production guards.
- Reconciled donor PRs #2–#8: no bulk adoption; reuse only named behavior after local authority, accessibility, and test review.
- Recorded the primary JSON-render path as complete, YAML/shadcn relationships as partial, and codegen/Ink/MCP/image/React Email/PDF/Three as non-product inventory.

G005 — Bounded reliability and first final gate

Started: 21:32:28Z · **Review-blocked:** 22:02:46Z

- Completed the staging-only GET health exercise and broad verification.
- Cleaner passed, but independent code review correctly blocked completion because the composed generate route did not pass the request-aware writer into both lifecycle instrumentation seams.
- A low-severity Markdown whitespace mismatch also invalidated an earlier broad `git diff --check` claim.
- The blocker appended G006 instead of being waived.

G006 — Persist stream lifecycle telemetry through the cloud writer

Started: 22:02:50Z · Review-blocked: 22:41:07Z

- Injected `writeRequestTelemetry` into spec-stream and generate-stream instrumentation.
- Proved owner-readable spec, tool, artifact, and terminal events and foreign-tenant isolation.
- Cleaner passed and independent code review approved.
- Architecture review still blocked one invariant: successful payloads could retain `reasoning`, `chainOfThought`, or `thinking` fields.
- The blocker appended G007.

G007 — Harden private-thought sanitization

Started: 22:41:19Z · Technical gates clear at: `030cdb8`

- Added the minimal shared-boundary filter in `sanitizeTelemetryValue`.
- Canonicalized key matching covers nested case, camelCase, snake_case, kebab-case, dotted, and spaced variants of reasoning, thinking, scratchpad, and chain-of-thought.
- Direct sanitizer and authenticated route-persistence regressions prove private sentinels are absent, safe siblings persist, secret redaction remains, and foreign scope is rejected.
- Cleaner: **PASS / no-op**. Code review: **APPROVE**. Architecture: **CLEAR**. Audit: **11/11 PASS**.

Ledger caveat: `.omx/ultragoal/goals.json` and `ledger.jsonl` still show G005 and G006 as `review_blocked` and G007 as `in_progress`. The technical blockers are closed by later evidence, but only the leader may write the final checkpoints and aggregate quality gate.

4. Verification and review evidence

Fresh product-head checks recorded by the final handoffs

GATE	RESULT	EVIDENCE
Agent observability build	PASS	Final leader verification at <code>030cdb8</code>
Workspace-session build	PASS	Final leader verification at <code>030cdb8</code>
Private-thought sanitizer regression	PASS	<code>agent-observability.test.mjs</code>
Authenticated telemetry persistence/isolation	PASS	<code>telemetry-route.test.mjs</code>

GATE	RESULT	EVIDENCE
Session rail and persistence-before-generate	PASS	<code>app-shell-session-rail.test.mjs</code>
Lifecycle/spec/tool cloud persistence	PASS	<code>workspace-cloud-sql-adapter.test.mjs</code>
Full typecheck	PASS	Exit 0; one known pre-existing Svelte warning
Agent UI production build	PASS	Final leader verification
Dev Workbench production build	PASS	Final leader verification
Whitespace/static diff	PASS	<code>git diff --check origin/main...HEAD</code>
Dependency/oRPC drift	PASS	No manifest/lockfile delta for G007; no oRPC introduced
Lint	N/A	Repository has no lint script/config; not misreported as PASS

Independent outcomes

The review sequence is part of the evidence, not noise to hide:

- 1. **G005 code review — REQUEST CHANGES.** Request-aware stream writer seams were not composed into the generate route.
- 2. **G006 code review — APPROVE.** The writer seam and owner/foreign persistence proof were repaired.
- 3. **G006 architecture review — BLOCK.** The generic sanitizer did not enforce the no-private-thought invariant.
- 4. **G007 cleaner — PASS / no-op.** The shared-boundary patch was already minimal.
- 5. **G007 code review — APPROVE.** No critical, high, medium, or low findings.
- 6. **G007 architecture review — CLEAR.** All 11 required invariants pass.

This sequence cost time, but it prevented the run from shipping durable lifecycle claims without composed persistence and prevented private-thought fields from entering logs or storage.

5. Bounded API reliability

Target: `GET https://sonik-booking-service-pipe-b.liam-trampota.workers.dev/api/v1/booking/ping`

Environment: staging **Bounds:** 5 virtual users for 30 seconds, GET only, no credentials, writes, fuzzing, production traffic, or Agent UI traffic.

MEASURE	RESULT
Requests / iterations	145 / 145
Checks	145 / 145 (100%)
Unexpected 5xx	0 / 145
HTTP request failures	0%
Average latency	53.468 ms

MEASURE	RESULT
Median latency	37.836 ms
p90 / p95	75.315 / 113.947 ms
Maximum latency	463.051 ms
Post-load recovery probe	HTTP 200

The result proves bounded endpoint liveness only. It does not prove authenticated downstream recovery, write paths, or production behavior. The checked-in OpenAPI fixture and live staging response also drift: staging added `membershipRole` and `organizationId`, resolved the credential-free request through a demo harness, and exposed an auth-posture mismatch for the nominally public ping operation. No scope widening or mutation was attempted.

6. Deployment receipts and operator validation

Agent UI

- URL: `https://sonik-agent-ui.liam-trampota.workers.dev`
- Worker version: `4308d685-cf6a-4f54-8480-85513b1d420f`
- `/api/version`: HTTP 200 with matching version and timestamp `2026-07-20T23:02:45.413158Z`

Dev Workbench

- Alias: `https://dev-workbench-sooty.vercel.app`
- Immutable URL: `https://dev-workbench-gj092pizs-danletterio-5975s-projects.vercel.app`
- Deployment ID: `dpl_FH7QGxZiPTw5FMZdjXBUwMpZLZhQ`
- Inspector: `https://vercel.com/danletterio-5975s-projects/dev-workbench/FH7QGxZiPTw5FMZdjXBUwMpZLZhQ`
- Anonymous alias probe: expected HTTP 401 with Basic realm `Sonik Dev Workbench`
- Anonymous immutable probe: HTTP 302 to Vercel authentication

Required operator pass

1. Confirm Agent UI `/api/version` matches the recorded Worker version.
2. Authenticate to the protected Workbench and record the deployment ID shown in the inspector.
3. From an authorized Booking host, open the embedded Workbench at narrow and desktop widths; verify source, context, layout, status, overflow, focus, and fullscreen exit controls.
4. Create or resume one session, submit one prompt, and record session/run/request correlation identifiers.
5. Wait for an explicit model/tool terminal state; verify spec/tool/finish telemetry is visible for the owning session and unavailable to a foreign tenant.
6. Reload, navigate away and back, and verify the same run and ordered history return without duplicate user messages or terminal sessions.

7. Exercise the authorized downstream workflow separately. Do not treat the Pipe B ping as recovery proof.

7. Pull request truth at report time

Remote PR 61 still reflects the pre-run head:

FIELD	REMOTE STATE
State	OPEN
Merge state	BLOCKED
Review decision	CHANGES_REQUESTED
Remote head	036f713f4f880b6e330f102a6fb687028dd9f9a9
Product/review head in this report	030cdb86eda3aedba326392f1ad4e363cf960b44
Old-head checks	Green, including types/tests/build, embedded smoke, Playwright E2E, API tripwire, malware guard, and CodeRabbit status

Green checks on 036f713 do not verify 030cdb8. The leader must push the final branch, wait for new-head CI, and obtain GitHub/CodeRabbit rereview before representing PR 61 as current or approved.

8. Insurance, rollback, and recovery

- **Small reversible fixes:** both late blockers were repaired at shared boundaries with focused regressions; no replacement framework or dependency was added.
- **Tests before behavior:** G002 pressure cases and G007 privacy coverage preserve the failing contract that motivated each fix.
- **Credential discipline:** inert sentinels test redaction; host authority and Cloudflare control-plane secrets do not enter guest sandbox paths.
- **Preview/staging first:** reliability traffic remained staging-only and read-only; deployments are protected operator targets, not an automatic production promotion.
- **Tripwire:** stop on unexpected 5xx, authority violation, credential exposure, or consequential mutation.
- **Known-good anchors:** baseline 036f713, product review head 030cdb8, immutable Workbench URL, deployment ID, and Agent UI Worker version provide rollback and comparison points.
- **Rollback mechanics:** revert the surgical branch commits and redeploy/promote the last known-good Worker or Vercel deployment. Do not live-patch credentials or bypass the protected host.
- **Data safety:** preserve one workspace/run identity, keep telemetry fail-soft, and use correlated Worker logs as fallback evidence when durable telemetry is unavailable.

9. Skills and process used

SURFACE	HOW IT WAS USED
Ultragoal	Durable brief, goal ledger, appended blocker stories, checkpoints, and aggregate closeout contract
OMX Team / worker lanes	Bounded implementation, verification, documentation, review, and deployment evidence handoffs
TDD and contract pressure testing	Six initial pressure groups plus focused session, writer-composition, privacy, and tenant-isolation regressions
API reliability testing	Safe GET-only staging exercise with explicit stop conditions and raw receipts
AI-slop cleaner	Changed-file review after behavior lock; G005/G006/G007 gates were PASS/no-op or documentation-only
Independent code review	Security, correctness, persistence, scoping, and evidence review across the branch delta
Independent architecture review	Eleven invariants covering continuity, privacy, persistence, sandbox authority, transport, theme, rendering, load safety, and dependencies
Sonik frontend/theme and accessibility lenses	Daisy/Sonik authority, semantic Callout repair, reduced-motion behavior, donor PR triage
Playwright/build/test tooling	Embedded journey coverage, production builds, targeted regressions, and document/PDF rendering

The working process was agreement-first, tests-first for behavioral changes, conservative at external boundaries, and evidence-driven at each checkpoint. The best outcome of the process was not the first green build; it was the willingness to append G006 and G007 when independent gates contradicted the completion narrative.

10. Time and token accounting

The accounting boundary begins at 2026-07-20T20:03:01Z; earlier branch work is excluded.

- Pre-close snapshot: **1,822,321 tokens**.
- Elapsed runtime at snapshot: **10,633 seconds (2h 57m 13s)**.
- Snapshot time: **2026-07-20 19:03 America/New_York**.
- Final artifact rendering, validation, commit, push, CI, rereview, and leader checkpoints add time and tokens after this snapshot.

These figures are intentionally labeled pre-close. They must not be combined with prior multi-day branch history or represented as the final total before the aggregate goal is closed.

11. Where the run lost efficiency

1. **The first plan was too broad.** The generator produced 18 goals before refinement to the five actual stories.

2. **Composed-route coverage arrived late.** G005 unit evidence proved the writer in isolation but not that the generate route supplied it to both instrumentation seams.
 3. **The privacy invariant was not executable early enough.** G006 repaired persistence before the architecture gate demonstrated that arbitrary reasoning-like keys survived the shared sanitizer.
 4. **Deployment preceded the final architecture gate.** Pre-G007 deployments had to be superseded after the privacy blocker was found.
 5. **Some checks were shape-sensitive.** The route-composition source assertion and Markdown whitespace mismatch produced avoidable confidence/maintenance friction.
 6. **Remote integration lagged local work.** The local branch accumulated the completed technical work while PR 61 stayed on `036f713`, leaving GitHub review and CI unable to validate the current head.
 7. **External proof authority was not available.** No authenticated operator credentials or authorized downstream `PROOF_URL` existed for the final live journey.
-

12. Improvements for the next targeted run

1. Refine the generated plan against explicit deliverables before starting the first implementation lane.
 2. Add one behavior-level route composition test at the same time as any injectable writer seam; do not rely on isolated helper coverage.
 3. Turn every security/architecture sentence into an executable invariant before the first final review, including canonical nested private-thought keys.
 4. Run code review and architecture review before deployment; deploy once after both are clear.
 5. Replace formatting-sensitive source assertions with behavior-level harnesses when the next route change touches them.
 6. Push at stable checkpoints so CI and reviewers assess the same head as local evidence.
 7. Pre-authorize protected login, Booking host context, and a safe downstream proof target before beginning the closeout lane.
 8. Keep reliability scope explicit: health, auth, workflow recovery, and writes are separate claims with separate permissions.
 9. Preserve the single Sonik theme and JSON-render catalog authority; adopt donor behavior one named gap at a time.
 10. Close the durable ledger immediately after each accepted repair so current goal state cannot lag final review evidence.
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13. Remaining gaps and closeout checklist

- ☐ Push the final branch, including this retrospective package.
- ☐ Confirm PR 61 head equals the pushed commit.
- ☐ Wait for new-head CI and CodeRabbit/GitHub rereview; clear the old `CHANGES_REQUESTED` state.
- ☐ Complete authenticated Workbench login and embedded Booking operator validation.
- ☐ Prove navigation/reload continuity and downstream workflow recovery in the authorized environment.
- ☐ Preserve the known Svelte initial-state warning and large client-chunk/Rollup `codeSplitting` warning as non-blocking follow-up, not hidden success.

-  Leader reconciles G005/G006/G007 in `.omx/ultragoal`, runs the final aggregate quality gate, and closes the aggregate Codex goal.
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14. Source evidence

- `.omx/context/final-ultragoal-retrospective-20260720T230342Z.md`
 - `.omx/ultragoal/brief.md`, `goals.json`, and `ledger.jsonl`
 - `docs/handoffs/sonik-agent-ui-completion-contract-2026-07-20.{md,html,pdf}`
 - `docs/handoffs/sonik-agent-ui-theme-json-render-audit-2026-07-20.md`
 - `.omx/handoffs/g005-{api-reliability,ai-slop-cleaner,independent-code-review}.md`
 - `.omx/handoffs/g006-{ai-slop-cleaner,independent-code-review,independent-architect-review,architecture-invariant-audit}.md`
 - `.omx/handoffs/g007-{ai-slop-cleaner,independent-code-review,independent-architect-review,architecture-invariant-audit}.md`
 - Live `gh pr view 61` snapshot and the current local/remote branch relation
-

15. Publication gate

This Markdown is the source for the self-contained HTML and printable PDF delivered beside it. Publication is complete only when all three files exist, the HTML has no external runtime assets, the PDF opens and renders with a non-zero page count and no observed clipping, content parity checks pass, hashes and sizes are recorded in the task completion evidence, `git diff --check` passes, and the three artifacts are committed together.